

Newton's record as a parallel move: what forgetting a record leaves

Result, 2026-09-27 (atlas of halving, open cell 6). Inserting an unobserved time into Newton's cell is a series move and closes exactly (refinement note, §2). Inserting a *record* at that time attaches a new variable, the pointer, to the body; forgetting it is the Newtonian counterpart of the parallel move of the series/parallel note. For a position record with Gaussian pointer of width σ this move is exact (Proposition 1): it leaves the body's position distribution unchanged and convolves its momentum distribution with a Gaussian of variance

$$v = \frac{\hbar^2}{4\sigma^2}.$$

Three consequences fill the Newton row of the atlas.

- **Records at one instant combine in parallel:** precisions add, $\sigma^{-2} = \sigma_1^{-2} + \sigma_2^{-2}$, and so do the momentum variances, as the heat times of parallel faces combine by conductances in the gauge case.
- **Refining with a record at every inserted time converges iff $\sum_j \sigma_j^{-2} < \infty$:** the total momentum diffusion is $\frac{\hbar^2}{4} \sum_j \sigma_j^{-2}$, and the same sum measures the information gathered about the place. The continuous-measurement limit of Caves and Milburn is the case $\sigma_j^{-2} = \kappa \tau_j$ (Corollary 2).
- **Information and disturbance are locked at $\hbar^2/4$ per unit:** $\sigma^2 v = \hbar^2/4$ for every Gaussian record and every mesh (for other pointers the product is bounded below by the Planck paper's Theorem 4). For $\hbar = 0$ the move is the identity for every σ on the classical phase-space distribution, since averaging the posterior over a forgotten outcome returns the prior (the factor $e^{-s^2/(8\sigma^2)}$ contains no $\hbar$; its effect on W is a momentum convolution of variance proportional to $\hbar^2$): records are series-exact, the Newtonian counterpart of a theory with no parallel planes, and exactly the joint determinacy of the fifth-postulate note.

The Gaussian record channel and its continuous limit are standard (Caves and Milburn 1987, metadata, cited in the Planck paper); the reading as the parallel move of the atlas is this note's. For non-Gaussian records the general floor is Theorem 6 of the Planck paper.

1. The record channel

A position record with Gaussian pointer of width σ and outcome y has Kraus operator M_y acting on wavefunctions by multiplication with $(2\pi\sigma^2)^{-1/4} e^{-(x-y)^2/(4\sigma^2)}$; forgetting the outcome gives the channel $\rho \mapsto \int M_y \rho M_y^\dagger dy$. Write $W(q, p)$ for the Wigner function of ρ .

Proposition 1. The forgotten record acts as $\rho(x, x') \mapsto \rho(x, x') e^{-(x-x')^2/(8\sigma^2)}$, that is,

$$W(q, p) \mapsto \int W(q, p - p') \frac{e^{-p'^2/(2v)}}{\sqrt{2\pi v}} dp', \quad v = \frac{\hbar^2}{4\sigma^2}.$$

The position marginal is unchanged; the momentum marginal is convolved with a centred Gaussian of variance v .

Proof. $\int M_y(x) M_y(x') dy = (2\pi\sigma^2)^{-1/2} \int e^{-[(x-y)^2 + (x'-y)^2]/(4\sigma^2)} dy = e^{-(x-x')^2/(8\sigma^2)}$. With $s = x - x'$ and $W(q, p) = (2\pi\hbar)^{-1} \int \rho(q + \frac{s}{2}, q - \frac{s}{2}) e^{-ips/\hbar} ds$, multiplication by $e^{-s^2/(8\sigma^2)}$ is convolution in p with the Gaussian whose characteristic function in $s/\hbar$ is $e^{-s^2/(8\sigma^2)}$, of variance v with $v/(2\hbar^2) = 1/(8\sigma^2)$. The factor equals one on the diagonal $s = 0$, so the position marginal is unchanged. $\square$

The record's error is σ and its delivered impulse has spread $\sqrt{v}$; their product is $\hbar/2$, the saturated mark cost of the Planck paper's Theorem 4.

2. Composition

Parallel. Two forgotten records at one instant multiply the density matrix by $e^{-s^2/(8\sigma_1^2)}e^{-s^2/(8\sigma_2^2)} = e^{-s^2(\sigma_1^{-2}+\sigma_2^{-2})/8}$: one record with $\sigma^{-2} = \sigma_1^{-2} + \sigma_2^{-2}$. Precisions add like conductances, and the momentum variances add. This is the Newtonian counterpart of two parallel faces at heat time $2t$ combining into one at t (the series/parallel note, §3).

Series. Between records at different times the body evolves freely or under constant force, an affine symplectic map on W ; a momentum convolution at one time becomes a sheared convolution in (q, p) at a later time. The channels compose exactly, because Gaussian convolutions and affine symplectic maps form a closed family.

Corollary 2 (refinement with records). Put a record of width σ_j at each vertex of a partition of $[0, T]$. The forgotten records convolve the final momentum distribution with a Gaussian of variance $\frac{\hbar^2}{4} \sum_j \sigma_j^{-2}$; the shear adds the position spread $\sum_j v_j(T - t_j)^2/M^2 \leq (T/M)^2 \sum_j v_j$. The noise covariance of the forgotten-record channel converges along the refinement iff $\sum_j \sigma_j^{-2}$ converges; for a divergent sum the momentum marginal flattens and the channel has no limit. The records' Fisher information about the initial place, for free motion with known momentum, is $\sum_j \sigma_j^{-2}$ and converges on the same condition. With $\sigma_j^{-2} = \kappa\tau_j$ the sum is κT : individual records become coarse as the mesh shrinks, and the rate κ and the momentum diffusion rate $\hbar^2\kappa/4$ survive the limit (the Lindblad term $-\frac{\kappa}{8}[x, [x, \rho]]$). This is a continuous limit of the kind Caves and Milburn construct; the Gaussian-pointer parametrization is stated here, their abstract being the part checked.

Proof. The multiplicative factors of Proposition 1 at one instant multiply; free evolution between vertices is an affine symplectic map, which carries Gaussian convolutions to Gaussian convolutions without changing the momentum variance of a convolution applied before it. The momentum variances therefore add (a harmonic force would rotate the kicks, which is why the corollary is stated for free and constant-force motion). $\square$

3. What the Newton row of the atlas now says

- **Series move** (unobserved insertion): exact after one scalar counterterm, at every $\hbar$.
- **Parallel move** (forgotten record): a momentum convolution of variance $\hbar^2/(4\sigma^2)$. It is the identity for $\hbar = 0$, as a gauge theory with no transverse planes has no parallel defect.
- **What survives refinement with records:** the information rate κ and the diffusion rate $\hbar^2\kappa/4$, with fixed product per record. This is the Newtonian form of the atlas's error budget: summable records, like summable parallel defects, give a limit, and the ratio that survives is set by the one action constant.

4. Consequence for STATE

Atlas cell 6 is filled for Gaussian records: forgetting a record is an exact parallel move whose defect is proportional to $\hbar^2$, which makes joint determinacy the statement that Newton's refinement has no parallel defect. The general, non-Gaussian record is covered by the Planck paper's Theorem 6 and by Theorem E of the fifth-postulate note.